

# **ASSIGNMENT II: MACHINE LEARNING MODELLING AND BIOLOGICAL DATA INTERPRETATION**

**10 JUNE 2026**

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**Weightage:** 10%

**Type:** By Group (2 – 3 students) , submit individually

**Duration:** 2–3 Hours (Can be completed during class)

**Tools:**

- Python
- Google Colab / Jupyter Notebook
- Scikit-learn
- Pandas
- Matplotlib / Seaborn

## **Dataset**

Students must use:

### **Option A (Recommended)**

The same dataset used in Assignment I.

OR

### **Option B**

A biological dataset approved by the lecturer:

Examples:

- Leukemia Gene Expression Dataset
- Breast Cancer Wisconsin Dataset
- Parkinson Disease Dataset
- Diabetes Dataset
- Heart Disease Dataset
- Gene Expression Omnibus (GEO) Dataset

### **Option C**

**From FYP 1 Dataset**

### **Task 1: Data Review (10 Marks)**

Briefly summarize:

- Number of samples
- Number of attributes
- Target class
- Data preprocessing performed in Assignment I

Provide:

- Dataset description table
- Screenshot/code snippet

### **Task 2: Machine Learning Model Implementation (30 Marks)**

Implement ONE machine learning algorithm:

#### **Classification Dataset**

Choose one:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- k-Nearest Neighbour (kNN)

#### **Regression Dataset**

Choose one:

- Linear Regression
- Random Forest Regressor

Students must:

1. Split dataset into training and testing sets
2. Train model
3. Generate predictions

### **Task 3: Model Optimization (20 Marks)**

Students must improve model performance using ONE optimization strategy:

Examples:

#### **Hyperparameter Tuning**

Examples:

- Number of neighbors (kNN)

- Max depth (Decision Tree)
- Number of trees (Random Forest)
- C value (SVM)

OR

### Feature Selection

Examples:

- SelectKBest
- RFE
- Correlation-based selection

Students compare:

| Model     | Before Optimization | After Optimization |
|-----------|---------------------|--------------------|
| Accuracy  |                     |                    |
| Precision |                     |                    |
| Recall    |                     |                    |

### Task 4: Performance Evaluation (20 Marks)

Students must calculate:

Classification:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

OR

Regression:

- MAE
- RMSE
- $R^2$

Include graphs where appropriate.

- *No screen shot*
- *Not directly from AI*

### Task 5: Biological Interpretation (20 Marks)

This is the most important part.

Students must answer:

### **Biological Meaning**

Examples:

- Which features contribute most to prediction?
- What biological significance do these features have?
- Are the results consistent with biological knowledge or literature?

Example:

*"Gene X showed high importance in leukemia classification, suggesting that abnormal expression of Gene X may be associated with disease progression."*

*Students should support interpretation using at least ONE scientific article.*

### **Deliverables**

Submit:

1. Python Notebook (.ipynb)
2. PDF Report (Maximum 5 pages)

### **Suggested Report Structure**

#### **1. Introduction**

Dataset and biological problem

#### **2. Methodology**

ML algorithm used

#### **3. Results**

Performance metrics

#### **4. Optimization**

Comparison before vs after optimization

#### **5. Biological Interpretation**

Meaning of findings

#### **6. Conclusion**

### **Marking Rubric (10%)**

| <b>Criteria</b> | <b>Marks</b> |
|-----------------|--------------|
| Dataset Review  | 10           |

| <b>Criteria</b>           | <b>Marks</b> |
|---------------------------|--------------|
| ML Implementation         | 30           |
| Optimization Technique    | 20           |
| Performance Evaluation    | 20           |
| Biological Interpretation | 20           |
| <b>Total</b>              | <b>100</b>   |

**Converted to 10% course assessment.**